

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

SEPTEMBER EXAMINATION

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS

SATURDAY, 28 SEPTEMBER 2024

TIME : 6.00 PM – 8.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Candidates are allowed to use scientific calculator.

Answer questions only in the answer booklet provided.

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**SECTION A**

- Q1. (a) Generate a smart plantation solution (architecture diagram) with Amazon Web Services (AWS) cloud platform. Justify the selection of each cloud service. Your design should consider:
- Deploying sensors for data capturing,
 - Storing data for plant growth analysis,
 - Generating serverless functions for data processing,
 - Notifying farmer for abnormal farm's conditions,
 - Having data pipeline for different cloud services,
 - Storing sensor log files into the cloud storage,
 - Visualizing sensor data for trend analysis, and
 - Integrating some relevant intelligent cloud services.
- (12 marks)
- (b) AWS offers several compute and database options to meet different needs. Which AWS service(s) can be used to host an off-premises MySQL database with minimal administrative operations? Justify your answer. (4 marks)
- (c) You have an Amazon Elastic Compute Cloud (EC2) Instance in development that interacts with the AWS DynamoDB. You will deploy the EC2 Instance in production environment soon. Which AWS service(s) should be used for secure communication between the EC2 Instance and the DynamoDB? Justify your answer. (3 marks)
- (d) Answer the following questions with short justification in the context of AWS.
- (i) Which AWS service would you use to monitor and manage the performance of your AWS resources and applications? (2 marks)
- (ii) Provide **TWO (2)** features of Amazon QuickSight that aid in deriving business value from its data. (2 marks)
- (iii) An external audit is being conducted on UTAR. The IT auditor requires a log of all accesses to the AWS resources in UTAR's account. Which AWS cloud service can assist in providing these details? (2 marks)
- [Total : 25 marks]
- Q2. (a) Compare and contrast Peer-to-Peer (P2P) architecture and Client-Server architecture. Discuss the key differences in terms of structure, performance, scalability, reliability, and use cases. (10 marks)

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**Q2. (Continued)**

- (b) Do you agree that the storage and processing of the data points collected by Internet of Things (IoT) devices will create more vulnerability? Provide **TWO (2)** reasons for your answer. (6 marks)
 - (c) In a distributed system scenario, under what circumstances would you choose message queuing over the publish/subscribe model? Please provide **TWO (2)** points to support your justification. (4 marks)
 - (d) Define microservices and present **TWO (2)** advantages of microservices over traditional web services. (5 marks)
- [Total : 25 marks]

- Q3. (a) Given a scenario of **4 Generals AND 1 Traitor**, find the traitor by using Byzantine Generals algorithm. Show all the steps and diagram necessary to your answer. (10 marks)
- (b) Under what condition can the Byzantine Generals Problem not be solved? (3 marks)
- (c) Four clocks W , X , Y and Z are performing internal synchronization by using Berkeley algorithm. W is reading 3:00:00.0 (hr:min:sec) when it is chosen as the master clock. At the same time, X , Y and Z are reading 2:50:00.0, 3:18:00.0 and 3:24:00.0, respectively. Illustrate the clock synchronization process with diagrams from initialization to adjust the time and show the calculations. (9 marks)
- (d) A clock, X , is reading 22:28:20.0 (hr:min:sec) when a clock synchronization request is sent to time server, S . After receiving the request from X , S prepares a response and appends the time 22:29:58.0 from its own clock. What will be the time set by X if the round-trip time (RTT) takes 8.0 seconds? Show your formula and calculation using Cristian's algorithm. (3 marks)
- [Total : 25 marks]

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**SECTION B**

- Q4. (a) Generate a smart car solution (architecture diagram) with AWS cloud platform. Your design should consider gathering of various data from the car and storing data into cloud storage and database for further analysis and visualization. (6 marks)
- (b) Recommend **ONE (1)** type of distributed computer system which is suitable to support 8 classes of students (40 students per class), where different classes required different types of mathematic/computer applications. Suppose that UTAR can only hire one lab administrator to take care of the machines. Provide **TWO (2)** justifications for your selection of distributed computer system. (6 marks)
- (c) Provide **THREE (3)** characteristics of cloud architecture that separate it from the traditional one. (6 marks)
- (d) Answer the following questions with short justification in the context of AWS.
- (i) What are the **TWO (2)** common AWS best practices that you should follow when designing and managing a secure cloud infrastructure on AWS? (4 marks)
- (ii) Which AWS service provides real-time guidance to help you provision your resources following AWS best practices, focusing on cost optimization, performance, security, fault tolerance, and service limits? (3 marks)
- [Total : 25 marks]
- Q5. (a) Explain the roles of each layer in an N-tier architecture. (6 marks)
- (b) Discuss **THREE (3)** benefits of migrating from a monolithic architecture to an N-tier architecture. (6 marks)
- (c) Elaborate **THREE (3)** key principles of microservices architecture that contribute to better application design and development practices? (6 marks)

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**Q5. (Continued)**

- (d) Which type of persistence communication can be best applied in transferring a patient's record from his admitted hospital *A* to his physician's hospital *B*? Describe the workflow for utilizing the selected communication service in the scenario above. *[Hints: Both hospitals are located in different cities, and aim to eliminate the traditional overhead associated with operating in-house communication infrastructures.]* (4 marks)
- (e) Describe a traffic shaping algorithm that can ensure a good quality of service during video streaming. (3 marks)

[Total : 25 marks]
